PRODUCT USER MANUAL

For Near-Real-Time Satellite Wave Products

WAVE_GLO_PHY_SWH_L3_NRT_014_001

WAVE_GLO_PHY_SWH_L4_NRT_014_003

WAVE_GLO_PHY_SPC_L3_NRT_014_009

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3.1	2024/05	All	New mapping method instantaneous SWH field in L4 SWH product Rename of 014_002, product and datasets	C.Béchonnet P. Zunino R. Husson	P. Zunino
4.0	2025/06	All	Product 014_002 and 014_004 removed from this document because they have their own PUM	P . Zunino	P. Zunino

Table of Contents

GLOSSARY AND ABBREVIATIONS *6*

DATA ACCESS **9**

I. INTRODUCTION *10*

Summary **10**

History of changes **11**

II DESCRIPTION OF THE PRODUCT SPECIFICATION *13*

II.1 General Information **13**

II.2 Details of datasets **16**

II.3 Production System Description **18**

II.3.1 WAVE-TAC SWH system **18**

II.3.2 WAVE-TAC SPC system **21**

II.4 Grid **24**

II.5 Processing information **24**

II.5.1 Update Time 24

II.5.1.1 WAVE-TAC SWH system 24

II.5.1.2 WAVE-TAC SPC system 24

II.5.2 Temporal extent of data stored on delivery mechanism 24

II.5.2.1 WAVE-TAC SWH system 24

II.5.2.2 WAVE-TAC SPC system 25

II.5.3 Time averaging 25

II.5.3.1 WAVE-TAC SWH system 25

II.5.3.2 WAVE-TAC SPC system 25

III FILE FORMAT *26*

IV FILES NOMENCLATURE *27*

IV.1 Nomenclature of files when downloaded through Marine Data Store **27**

IV.2 Nomenclature of original files **27**

IV.3 Other information: mean centre of Products, land mask value, missing value **28**

IV.4 File size **29**

IV.5 Structure of files **29**

V REFERENCES *30*

VI ANNEX *31*

VI.1 SWH level-3 files (_014_001) **31**

VI.2 VI.2 SWH level-4 files (_014_003) **32**

VI.3 SPC OBS CFOSAT level-3 files (_014_009) **35**

GLOSSARY AND ABBREVIATIONS

ADD	Architecture and Design Document
AL	Saral/AltiKa
C2	Cryosat-2
CalVal	Calibration/Validation
CF	Climate Forecast (convention for NetCDF)
CFO	CFOSAT
CLS	Collecte Localisation Satellite
CMEMS	Copernicus Marine Environment Monitoring Service
DT	Delayed Time
DU	Dissemination Unit
GDR	Geophysical Data Record
GOP	Geophysical Ocean Product
HY-2B	Hai Yang 2B
HY-2C	Hai Yang 2C
IGDR	Intermediate Geophysical Data Record
J3	Jason-3
KPI	Key Production Indicator
L2/L2P/L3/L4	Level 2 / Level 2 Plus / Level 3 / Level 4
LRM	Low Resolution Mode
MOTU	Web server allowing the distribution of met/ocean gridded data files through the web (https://github.com/clstoulouse/motu)
NetCDF	Network Common Data Form
NOP	NRT Ocean Product

NRT	Near Real Time
OE	Orbit Error
OGDR	Operational Geophysical data record
OLA	Operational Level Agreement
PIT	Product Information Table
PLRM	Pseudo LRM
PQ	Product Quality
PU	Production Unit
PUM	Product User Manual
QUID	QUality Information Document
RMS	Root Mean Square
RPC	Remote Protocol Client
RT	Real Time
S1A	Sentinel-1A
S1B	Sentinel-1B
S3A	Sentinel-3A
S3B	Sentinel-3B
S6A	Sentinel-6A
SAR	Synthetic Aperture Radar
SARIn	SAR Interferometric
SPC	SPeCtral
SRD	System Requirement Document
SWH	Significant Wave Height
SWON	SWOT nadir
TAC	Thematic Assembly Center
TDS	Thredds Data Server

VAVH	Visual AVerage Height
WMS	Web Map Service

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If you have any questions, please contact us.

I. INTRODUCTION

This document is the user manual for the Copernicus Marine Service Near-Real-Time satellite wave products produced by the WAVE-TAC (Thematic Assembly Centre).

Summary

The WAVE TAC (Thematic Assembly Centre) is one of the eight TAC of the Copernicus Marine Service. The data produced in the frame of this TAC are generated by different processing systems including data from altimeter and other wave measuring devices (SAR, SWIM) for all the Copernicus missions (Sentinel-6A, Sentinel-3A/B, Sentinel-1A/B) and other collaborative or opportunity missions (e.g. : Jason-3, HaiYang-2 missions, Saral/AltiKa, Cryosat-2, CFOSAT).

The WAVE-TAC product lines described here consist of Near Real Time (NRT) Level-3 (L3; along-track) and Level-4 (L4; gridded) products. Another Product user manual (PUM) is dedicated to WAVE-TAC Delayed Time/ Multi-Year (DT/MY) products. Through the WAVE-TAC, the Copernicus Marine Service also disseminates L3 products/datasets produced under Eumetsat responsibility. They mainly concern the Copernicus altimeter missions (Sentinel-3 & Sentinel-6).

The following Level-3 and Level-4 products are delivered in Near-Real-Time by the WAVE-TAC and are detailed in this user manual:

- WAVE_GLO_PHY_SWH_L3_NRT_014_001: altimetry along-track Significant Wave Height (SWH) level-3 product based on Sentinel-6A, Jason-3, Sentinel-3A, Sentinel-3B, Saral/AltiKa, Cryosat-2, CFOSAT nadir, SWOT nadir, Hai Yang-2B and Hai Yang-2C measurements.
- WAVE_GLO_PHY SPC_L3_NRT_014_009: Directional wave spectra and associated integrated parameters (Hs, direction and wavelength for the entire spectrum and for wave partitions) level-3 product based on rotating Radar SWIM off-nadir measurements on-board CFOSAT mission.
- WAVE_GLO_PHY_SWH_L4_NRT_014_003: Gridded significant wave height level-4 product based on all available significant wave height level-3 products.

History of changes

Date	Description of changes and impacted product
DEC-2020	Quality improvement of the mapping of the variable VAVH_INST in the product WAVE_GLO_WAV_L4_SWH_NRT_OBSERVATIONS_014_003
DEC-2021	New gridded product WAVE_GLO_PHY_SPC_L4_NRT_014_004 merging spectral wave measurements issued from Sentinel-1A and -1B missions. New dataset cmems_obs-wave_glo_phy-spc_nrt_cfo-l3_PT10S within the WAVE_GLO_WAV_L3_SPC_NRT_OBSERVATIONS_014_002 product, which contains swell system integral parameters measurements provided by CFOSAT SWIM off-nadir instrument in Near-Real-Time and for global area.
APR-2022	New dataset cmems_obs-wave_glo_phy-swh_nrt_s6a-l3_PT1S within the WAVE_GLO_WAV_L3_SWH_NRT_OBSERVATIONS_014_001 product, containing Sentinel-6A altimeter LRM measurements Change of orbit of Jason-3 (new orbit expected in June 2022)
JUN-2022	Change of name for L3 and L4 SWH products: WAVE_GLO_WAV_L3_SWH_NRT_OBSERVATIONS_014_001→ WAVE_GLO_PHY_SWH_L3_NRT_014_001 WAVE_GLO_WAV_L4_SWH_NRT_OBSERVATIONS_014_003→ WAVE_GLO_PHY_SWH_L4_NRT_014_003 Dataset names changed New dataset for HY-2C: cmems_obs-wave_glo_phy-swh_nrt_h2c-l3_PT1S Improvement in data quality for both L3 and L4 SWH with new lookup tables
JUN-2023	New dataset for SWOT nadir: cmems_obs-wave_glo_phy-swh_nrt_swon-l3_PT1S
JAN-2024	Update product WAVE_GLO_PHY_SPC_L4_NRT_014_004, merging Sentinel-1A and CFOSAT missions (change carried out in the EIS November 2023. Addition of a new L3 SPC NRT
MAY-2024	Upate product WAVE_GLO_PHY_SWH_L4_NRT_014_003: new mapping method for the instantaneous swh field.

	Rename 014_002 product name and dataset names

II DESCRIPTION OF THE PRODUCT SPECIFICATION

II.1 General Information

Product Lines	WAVE_GLO_PHY_SWH_L3_NRT_014_001
Geographical coverage	Global Ocean
Variables	**VAVH**: sea surface significant wave height, i.e. the mean wave height of the highest third of the waves (H1/3) **VAVH_UNFILTERED**: sea surface significant wave height, i.e. the mean wave height of the highest third of the waves (H1/3), before applying the along-track denoising **WIND_SPEED**: 10-meter wind speed
Product Type	Near-Real-Time Level-3 observations
Available time series	JAN-2020 – Present (depends on mission, see QUID)
Temporal resolution	Instantaneous (one file contains 3 hours of measurements)
Target delivery time	3 hours after the acquisition of upstream data (L2P)
Delivery mechanism	Marine Data Store
Horizontal resolution	7km (along-track)
Number of vertical levels	1 (Surface only)
Format	NetCDF-4

Table 1: WAVE-TAC Near-Real-Time WAVE_GLO_PHY_SWH_L3_NRT_014_001 product

Product Lines	WAVE_GLO_PHY_SWH_L4_NRT_014_003
Geographical coverage	Global Ocean

Variables	**VAVH_DAILY_MEAN**: daily mean significant wave height **VAVH_DAILY_STD**: daily standard deviation of significant wave height **VAVH_DAILY_MAX**: daily maximum significant wave height **VAVH_DAILY_NBR**: daily number of significant wave height measurements **VAVH_INST**: instantaneous significant wave height estimated from all available satellite measurements, accounting for their spatial and temporal proximity. **VAVH_INST_FLAG**: flag associated with a post-processing modification of the data (threshold set to 0 m). Where the value was invalid, the flag is set to 1, otherwise to 0.
Product Type	Near-Real-Time Level-4 observations
Available time series	JAN 2021 – Present
Temporal resolution	Daily
Target delivery time	Daily: 12:00 UTC
Delivery mechanism	Marine Data Store
Horizontal resolution	2° x 2°
Number of vertical levels	1 (Surface only)
Format	NetCDF-4

Table 3: WAVE-TAC Near-Real-Time WAVE_GLO_PHY_SWH_L4_NRT_014_003 product

Product Lines	WAVE_GLO_WAV_L3_SPC_NRT_014_009
Geographical coverage	Global Ocean

Variables	This product contains, for every along track data, all the spectral observations (wave spectrum and all integrated parameters, including swells and wind wave, partitioned). Unlike fireworks, it does not contain any propagation of swell but only observations. **Wave_spec** = Estimated directional wave spectrum for spectral beam 10 degrees **Wave_spec_partition_mask** = Mask of the partitions detected on the wave spectrum (1 and -1 indicate the partitions and its symmetric counterpart, 0 is outside partition) **VAVH =** SWH estimated from whole spectrum. **VAVH_partition** = SWH estimated for the three dominant partitions of the spectrum **VTPK =** Peak period estimated from the whole spectrum. **VTPK_partition** = Peak period estimated for the three dominant partitions of the spectrum. **VPED** = Peak direction estimated from the whole spectrum. It corresponds to the wave direction (with 180 deg. ambiguity) in degrees from North, clockwise. **VPED_partition** = Peak direction estimated for the three dominant partitions of the spectrum" Where invalid, these variables are replaced by their fill values.
Product Type	Near-Real-Time Level-3 wave spectra
Available time series	APR-2024 – Present (see QUID)
Temporal resolution	Instantaneous (one file contains 3 hours of measurements)
Target delivery time	3 hours after the acquisition of upstream data (L2P)
Delivery mechanism	Marine Data Store
Horizontal resolution	Discrete

Number of vertical levels	1 (Surface only)
Format	NetCDF-4

Table 5: WAVE-TAC Near-Real-Time WAVE_GLO_PHY_SPC_L3_NRT_014_009 product

II.2 Details of datasets

WAVE_GLO_PHY_SWH_L3_NRT_014_001
Datasets (one per mission): cmems_obs-wave_glo_phy-swh_nrt_al-l3_PT1S cmems_obs-wave_glo_phy-swh_nrt_c2-l3_PT1S cmems_obs-wave_glo_phy-swh_nrt_cfo-l3_PT1S cmems_obs-wave_glo_phy-swh_nrt_h2b-l3_PT1S cmems_obs-wave_glo_phy-swh_nrt_h2c-l3_PT1S cmems_obs-wave_glo_phy-swh_nrt_j3-l3_PT1S cmems_obs-wave_glo_phy-swh_nrt_s3a-l3_PT1S cmems_obs-wave_glo_phy-swh_nrt_s3b-l3_PT1S cmems_obs-wave_glo_phy-swh_nrt_s6a-l3_PT1S cmems_obs-wave_glo_phy-swh_nrt_swon-l3_PT1S
VAVH [m] Significant wave height (H1/3) on main altimeter frequency band sea_surface_wave_significant_height
VAVH_UNFILTERED [m] Significant wave height (H1/3) on main altimeter frequency band sea_surface_wave_significant_height
WIND_SPEED [m s-1] Equivalent 10-meter wind speed (U10) derived from altimeter measurements wind_speed

Table 6: list of the datasets and variable names and unit for the WAVE_GLO_PHY_SWH_L3_NRT_014_001 product

WAVE_GLO_PHY_SWH_L4_NRT_014_003
Dataset (one for instantaneous SWH and one for daily mean): **cmems_obs-wave_glo_phy-swh_nrt_multi_l4_2deg_P1D-m** **cmems_obs-wave_glo_phy-swh_nrt_multi_l4_2deg_P1D-i**

VAVH_DAILY_MEAN [m] Daily mean significant wave height sea_surface_wave_significant_height daily_mean
VAVH_DAILY_STD [m] Daily standard deviation of significant wave height sea_surface_wave_significant_height daily_standard_deviation
VAVH_DAILY_MAX [m] Daily maximum significant wave height sea_surface_wave_significant_height daily_maximum
VAVH_DAILY_NBR [-] Daily number of significant wave height measurements sea_surface_wave_significant_height daily_number_of_observations
VAVH_INST [m] Instantaneous significant wave height sea_surface_wave_significant_height
VAVH_INST_FLAG [-] Flag associated with the modification of VAVH_INST variable. sea_surface_wave_significant_height_flag

Table 2: list of the datasets and variable names and unit for the WAVE_GLO_PHY_SWH_L4_NRT_014_003 product

WAVE_GLO_PHY_SPC_L3_NRT_014_009
Dataset: cnes_obs-wave_glo_phy-spc_nrt_cfo-l3_PT10S
dimensions: **side_looking (0 or 1)** **obs stands for the along track position (every 70km)** **spectrum_wavenumber stands for the polar indice of the 2D spectrum** **spectrum_direction (24 stands for the angular resolution)** **X = n_partition = stands for the number of the partition (0 to 2)** ***2D spectra part:*** **wavenumber_spec(spectrum_wavenumber):** Wavenumbers for polar spectra (in rad/m) **direction_spectrum(spectrum_direction):** Array of angular values for polar spectra (degrees from North, clockwise) **Wave_spec**(spectrum_wavenumber, spectrum_direction, side_looking, obs) = Estimated directional wave spectrum for spectral beam 10 degrees

Wave_spec_partition_mask(spectrum_wavenumber, spectrum_direction, side_looking, obs) = Mask of the partitions detected on the wave spectrum (1 and -1 indicate the partitions and its symmetric counterpart, 0 is outside partition) ***Integrated parameters part:*** **VAVH[m] =** SWH estimated from whole spectrum. **VAVH_partition[X][m]** = SWH estimated for the three dominant partitions of the spectrum **VTPK[s] =** Peak period estimated from the whole spectrum. **VTPK_partition[X][s]** = Peak period estimated for the three dominant partitions of the spectrum. **VPED[deg]** = Peak direction estimated from the whole spectrum. It corresponds to the wave direction (with 180 deg. ambiguity) in degrees from North, clockwise. **VPED_partition[X][deg] = Peak direction estimated for the three dominant partitions of the spectrum**

Table 9: list of the datasets and variable names and unit for the WAVE_GLO_PHY_SPC_L3_NRT_014_009 product

Note for users: The VAVH variable stands for Visual AVerage Heigh of the highest 1/3 wave (H1/3). Significant wave height is a statistic computed from wave measurements and corresponds to the average height of the highest one third of the waves, where the height is defined as the vertical distance from a wave trough to the following wave crest. The first 'V' in VAVH comes from the in-situ community.

II.3 Production System Description

II.3.1 WAVE-TAC SWH system

WAVE_GLO_PHY_SWH_L3_NRT_014_001

Summary of the operational configuration:

Domain and resolution: The altimeter measurements are delivered over global Ocean. The measurements are made along-track at the surface with a 7-km resolution.

Geographic coverage: For Jason-3, Sentinel-6A and Hai Yang-2C, the coverage is maximum 66°S/66°N in latitude; for Sentinel-3A, Sentinel-3B and AltiKa, 81.5°S/81.5°N; for Cryosat-2, 88°S/88°N; for HaiYang-2B, 81°S/81°N; for CFOSAT, 83°S/83°N and for SWOT, 78°S/78°N. Note that for Sentinel-3A and B, the altimeter wave measurements are deduced from SAR measurements. See QUID for details.

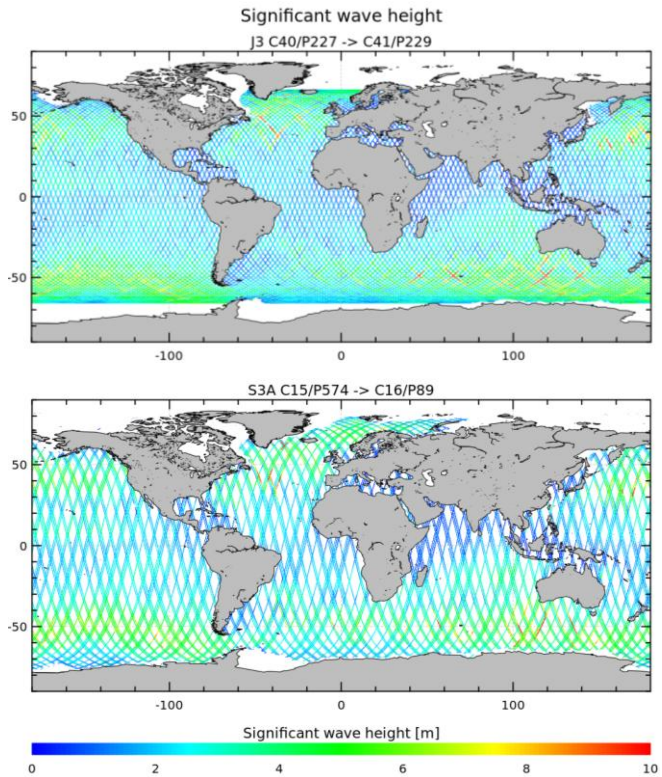

Figure 1*: Example of wave measurements from along-track Jason-3 (top) and Sentinel-3A (bottom) altimeters. One point every 7km along the satellite passes. The coverage is 10 days [pass 227 cycle 40 to pass 229 cycle 41 (one cycle is 10 days)] for Jason-3 and [pass 574 cycle 15 to pass 89 cycle 16 (one cycle is 27 days)] for Sentinel-3A*

WAVE_GLO_PHY_SWH_L4_NRT_014_003

Summary of the operational configuration:

Domain and resolution: The gridded product merges all available Significant Wave Height measurements over the global ocean. The maps are delivered daily at a 2-degree resolution.

Geographic coverage: The geographic coverage is constrained by the availability of wave measurements (see the geographic coverage for Level-3 SWH product above) and by the sea ice concentration (updated daily from Copernicus Marine Service SEAICE_GLO_SEAICE_L4_NRT_OBSERVATIONS_011_001 product).

The VAVH_INST fields utilise wave measurements, accounting for their spatial and temporal proximity to provide wave information over each open ocean grid cell. The VAVH_DAILY fields are computed from the measurements available over one day (from 00UTC to 23:59UTC), filling only grid cells with measurements. See QUID for details.

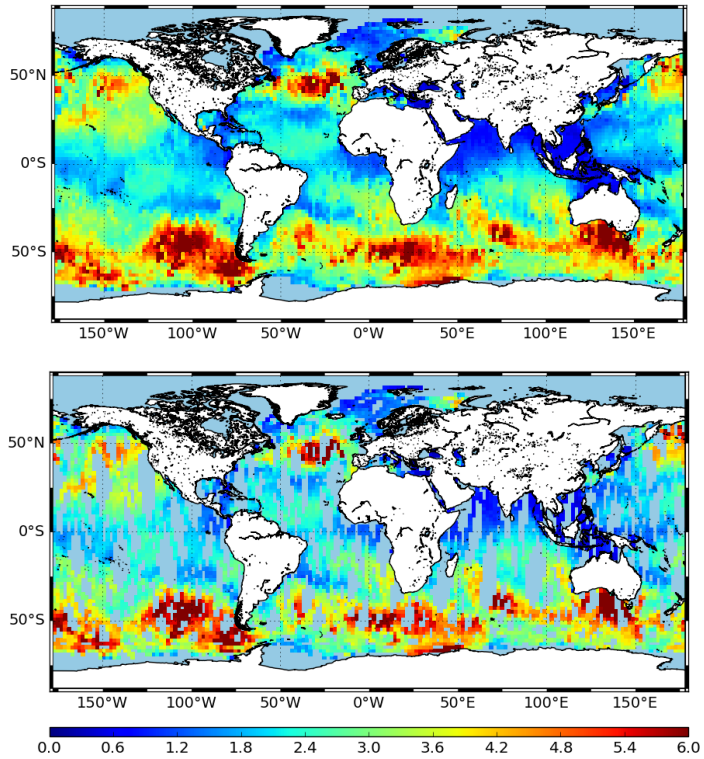

Figure 2: *Example of gridded level-4 significant wave height maps for the VAVH_INST field (top), estimated from all available missions Level 3 along-track measurements, accounting for their spatial and temporal proximity and the VAVH_DAILY_MEAN field (bottom), computed from daily measurements.*

II.3.2 WAVE-TAC SPC system

WAVE_GLO_PHY_SPC_L3_NRT_014_009
Summary of the operational configuration: **Domain and resolution:** CFOSAT spectra measurements are delivered over global Ocean. The measurements are made along-track at the surface with a 70-km resolution, apart from the nadir, both on the left and on the right of the track, 45km apart over boxes of 70km/90km. **Geographic coverage**: The highest latitudes covered are CFOSAT, 83°S/83°N. See QUID for details.

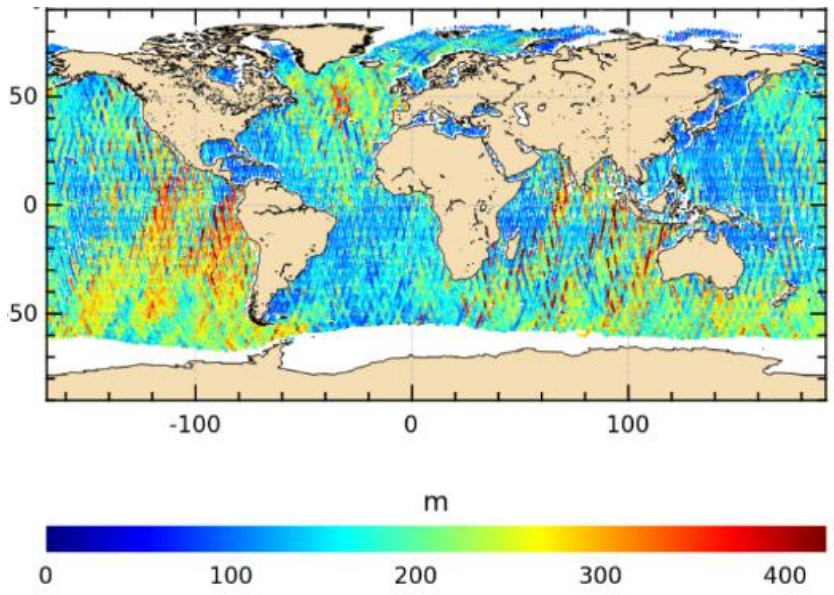

Figure 4*: Example of peak wavelength (in meters) measurements from along-track off nadir CFOSAT (left side). One point every 70km along the satellite passes. The repeat cycles last 13 days.*

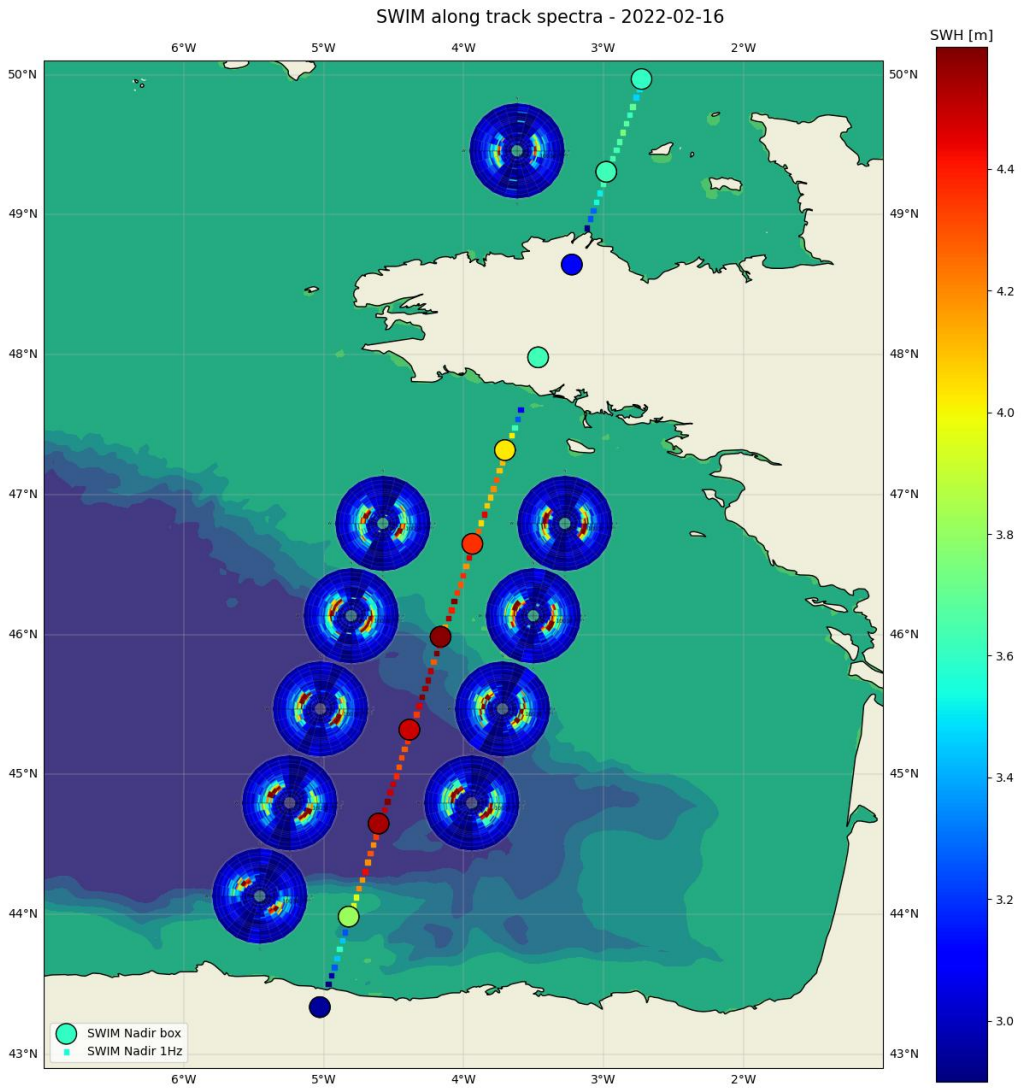

Figure 5*: Example Left/right sampling apart from the nadir (SWH, dotted colored points) near Britany (France).*

See the QUID report for complete information about the processing.

II.4 Grid

All Level-4 gridded products are delivered onto cartesian grids. The resolution of the grid varies according to the product considered. All the variables are defined at the center of the output grid cell.

II.5 Processing information

II.5.1 Update Time

II.5.1.1 WAVE-TAC SWH system

The production chain runs every 30 minutes and creates or updates L3 files if new upstream data are acquired. L3 products are formatted into 3-hourly files (starting at 00, 03, 06, 09, 12, 15, 18 and 21UTC). As soon as L2P data are acquired within a 3-hour time window, a new L3 file is generated. This file is then updated every 30 minutes, as long as new L2P data are acquired within the 3-hour time window, increasing the temporal coverage.

In Copernicus Marine catalogue, a purge is performed at each delivery in order to delete the oldest and least complete file for each 3-hour time window. However, the user can also keep track of the successive updates, using the production date included in the file name.

The production chain runs once a day to produce the day-1 and update up to day-5 level-4 files by 12:00 UTC. Each daily run incorporates the latest available L3 measurements.

II.5.1.2 WAVE-TAC SPC system

In Copernicus Marine catalogue, a purge is performed at each delivery in order to delete the oldest and least complete files for all L3 products. However, the user can also keep track of the successive updates, using the production date included in the file name.

The production chain runs once a day to produce level-3 SPC products from all latest available L2 measurements. The L4 production is then run from all available L3 products.

Level-3 and level-4 files are updated by 12:00 UTC.

II.5.2 Temporal extent of data stored on delivery mechanism

II.5.2.1 WAVE-TAC SWH system

At each delivery, the significant wave height INST and DAILY fields are computed for day-1, day-2 and day-5. The day-1 maps are created as a new file whereas the day-2 and day-5 maps are updated using a more complete and up-to-date set of observations.

II.5.2.2 WAVE-TAC SPC system

At each delivery, the swell field analysis is performed over the last 3 weeks. This is to take into account the possible swell propagation from the storm source for more than 10 days before reaching the coast plus a potential storm duration of several days.

After this swell field analysis, the older swell fields are almost unchanged from one update to the next as very few observations can belong to the oldest swell fields. For the more recent days, new swell fields can appear from one day to the next if enough L2 observations have been gathered to create a new swell system. Also, older swell fields can be enriched with additional recent L2 observations.

All L4 products covering the swell field analysis time period (the last 3 weeks) are updated in this process.

II.5.3 Time averaging

II.5.3.1 WAVE-TAC SWH system

Instantaneous values are provided in L3 files (time is specified in the netCDF files). There is no temporal averaging nor interpolation performed, a low pass filter is applied to remove high frequency noise (see QUID for details).

In L4 files, an instantaneous value VAVH_INST is estimated from L3 measurements available within a time window around the date of the map (12 UTC). Daily statistics are also provided in the VAVH_DAILY fields (mean, standard deviation, maximum and number of measurements) and are computed using level 3 along-track measurements of the day from 00 UTC until 23:59 UTC.

II.5.3.2 WAVE-TAC SPC system

Instantaneous values are provided. There is no temporal averaging or interpolation performed.

III FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:

What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:

how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

IV FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this article: " How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center "

IV.1 Nomenclature of files when downloaded through Marine Data Store

Only gridded level-4 products are available through the Copernicus Marine Service Web Portal .

Files nomenclature when downloaded through the Copernicus Marine Web Portal is based on product dataset name and a numerical reference related to the request date on the MIS.

The scheme is: **datasetname_nnnnnn.nc**

where:

datasetname is a character string within one of the datasets listed in section 0

nnnnnn: 6-digit integer corresponding to the current time (download time) in milliseconds since January 1, 1970, midnight UTC.

nc: standard NetCDF filename extension.

Example:

cmems_obs-wave_glo_phy-swh_nrt_s6a-l3_PT1S_435283.nc

Specific case:

As the L4 SPC product (WAVE_GLO_PHY_SPC_L4_NRT_014_004) contains char variables in storm_name, this variable cannot be visible nor extractible or visualized from THREDDS (WMS) and MOTU (subset). Yet, it is however available in the original files distributed on Marine Data Store.

IV.2 Nomenclature of original files

All WAVE-TAC products (along-track level-3 and gridded level-4) are available through the Marine Data Store.

The nomenclature of files is described below for the different wave products.

- WAVE_GLO_PHY_SWH_L3_NRT_014_001 filenames:

global_vavh_l3_rt_<mission>_<datestart>_< dateend>_<dateprod>.nc

With:

- <mission>= s3a, s3b, j3, al, c2, cfo, h2b, h2c, or s6a_lr
- < datestart >= date of the start of the 3-hour period YYYYmmddTHHMMSS
- <dateend>= date of the end of the 3-hour period YYYYmmddTHHMMSS
- <dateprod>=date of production of the file YYYYmmddTHHMMSS

- WAVE_GLO_PHY_SWH_L4_NRT_014_003 filenames:

cmems_obs-wave_glo_phy-swh_nrt_multi_l4_2deg_P1D-<subfield>_< date>_P<dateprod>.nc

With:

- <subfield>= i for instantaneous and m for dailymean
- < date>= date of the daily wave field YYYYmmddT120000Z
- <dateprod>=date of production of the file YYYYmmddTHHMMSSZ

- WAVE_GLO_PHY_SPC_L3_NRT_014_009 filenames:

cnes_obs-wave_glo_phy-spc_nrt_cfo-l3_PT10S_<datemin>_<datemax>_P<productiontime>.nc

With:

- <mission>= cfo
- < datemin >= date of first measurement in the file YYYYmmddTHHMMZ
- <datemax>= date of last measurement in the file YYYYmmddTHHMMZ
- < productiontime >= date of production of the file YYYYmmddTHHMMZ

IV.3 Other information: mean centre of Products, land mask value, missing value

For some parameters, the NetCDF format is used with short integer coding for best compression, using an offset and scale factor as shown below:

Real_Value = (Display_Value X scale_factor) + add_offset

Missing value, valid_min and valid_max values are to be compared to the raw netcdf data, before applying the scale_factor and add_offset.

The missing value for the SWH products is: -32767s (-999 for the number of measurements).

The missing value for the SPC products is: -999.

For gridded level-4 products, land and sea-ice masks are equal to "_FillValue".

For all variables, missing values, scale factor and offset are given in netCDF variable attributes.

IV.4 File size

Product type	Product level	NetCDF file size
SWH	**L3**	Up to 200 Ko
	L4	100 Ko the -m files and 1MB the -i files
SPC OBS CFOSAT	**L3**	Less than 6 Mo per file

IV.5 Structure of files

Examples of the header of output NetCDF files are inserted in annex, for each dataset.

V REFERENCES

QUID: the associated QUID documents for each product are CMEMS-WAVE-QUID-014-001, CMEMS-WAVE-QUID-014-002, CMEMS-WAVE-QUID-014-003, CMEMS-WAVE-QUID-014-004 and CMEMS-WAVE-QUID-014-009.

VI ANNEX

VI.1 SWH level-3 files (_014_001)

Example of instantaneous NetCDF metadata from these files is given below:

```
netcdf global_vavh_l3_rt_s3a_20190801T210000_20190802T000000_20190830T112509 {
dimensions:
        time = 6689 ;
variables:
        double time(time) ;
                time:long_name = "time (sec. since 2000-01-01)" ;
                time:standard_name = "time" ;
                time:units = "seconds since 2000-01-01 00:00:00.0" ;
                time:calendar = "gregorian" ;
                time:axis = "T" ;
        int latitude(time) ;
                latitude:scale_factor = 1.e-06 ;
                latitude:valid_min = -90000000 ;
                latitude:comments = "Positive latitude is North latitude, negative latitude is South latitude." ;
                latitude:long_name = "latitude" ;
                latitude:standard_name = "latitude" ;
                latitude:units = "degrees_north" ;
                latitude:valid_max = 90000000 ;
        int longitude(time) ;
                longitude:scale_factor = 1.e-06 ;
                longitude:valid_min = 0 ;
                longitude:comments = "East longitude relative to Greenwich meridian" ;
                longitude:long_name = "longitude" ;
                longitude:standard_name = "longitude" ;
                longitude:units = "degrees_east" ;
                longitude:valid_max = 360000000 ;
        short VAVH(time) ;
                VAVH:_FillValue = -32767s ;
                VAVH:comment = "Bias corrected and filtered. Calibration relative to buoys [Sepulveda et al, 2015]" ;
                VAVH:scale_factor = 0.001 ;
                VAVH:valid_min = 0s ;
                VAVH:coordinates = "longitude latitude" ;
                VAVH:long_name = "Significant Wave Height on main altimeter frequency band" ;
                VAVH:standard_name = "sea_surface_wave_significant_height" ;
                VAVH:units = "m" ;
                VAVH:valid_max = 32767s ;
        short VAVH_UNFILTERED(time) ;
                VAVH_UNFILTERED:_FillValue = -32767s ;
                VAVH_UNFILTERED:comment = "Bias corrected. Calibration relative to buoys [Sepulveda et al, 2015]" ;
                VAVH_UNFILTERED:scale_factor = 0.001 ;
                VAVH_UNFILTERED:valid_min = 0s ;
                VAVH_UNFILTERED:coordinates = "longitude latitude" ;
                VAVH_UNFILTERED:long_name = "Significant Wave Height on main altimeter frequency band" ;
                VAVH_UNFILTERED:standard_name = "sea_surface_wave_significant_height" ;
                VAVH_UNFILTERED:units = "m" ;
                VAVH_UNFILTERED:valid_max = 32767s ;
```

```
        short WIND_SPEED(time) ;
                WIND_SPEED:_FillValue = -32767s ;
                WIND_SPEED:comment = "Bias corrected relative to a reference mission" ;
                WIND_SPEED:valid_min = 0s ;
                WIND_SPEED:valid_max = 32767s;
                WIND_SPEED:coordinates = "longitude latitude" ;
                WIND_SPEED:long_name = "Equivalent 10-m wind speed derived from altimeter measurements";
                WIND_SPEED:standard_name = "wind_speed" ;
                WIND_SPEED:units = "m s-1" ;
                WIND_SPEED:scale_factor = 0.001 ;
// global attributes:
                :Conventions = "CF-1.6" ;
                :first_meas_time = "2019-08-01 21:01:17" ;
                :last_meas_time = "2019-08-01 23:59:59" ;
                :comment = "Significant Wave Height measured by altimetry" ;
                :keywords_vocabulary = "NetCDF COARDS Climate and Forecast Standard Names" ;
                :creator_email = "servicedesk.cmems@mercator-ocean.eu" ;
                :references = "http://marine.copernicus.eu" ;
                :platform = "Sentinel-3A" ;
                :Metadata_Conventions = "Unidata Dataset Discovery v1.0" ;
                :keywords = "Oceans > Ocean Topography > significant wave height" ;
                :institution = "CLS, CNES, EUMETSAT" ;
                :creator_name = "CMEMS - Wave Thematic Assembly Center" ;
                :license = "http://marine.copernicus.eu/web/27-service-commitments-and-licence.php" ;
                :title = "NRT Sentinel-3A Global Ocean Along track significant wave height L3 product" ;
                :standard_name_vocabulary = "NetCDF Climate and Forecast (CF) Metadata Convention Standard Name
Table v37" ;
                :project = "EUMETSAT Sentinel-3 L3 Wave Marine Altimetry Service" ;
                :cdm_data_type = "swath" ;
                :contact = "servicedesk.cmems@mercator-ocean.eu" ;
                :source = "Sentinel-3A measurements" ;
                :based_on = "L2P Sentinel-3A NRT" ;
                :creator_url = "http://marine.copernicus.eu" ;
                :processing_level = "L3" ;
                :equator_time = "2019-08-01T20:46:34.461000" ;
                :equator_longitude = 18.39 ;
                :creation_date = "2019-08-30T11:25:09" ;
                :product_version = "2_0" ;
                :software_version = "L3RT_SWH: 1.2.0-SNAPSHOT; OCTANT: 12.18.0" ;
                :history = "2019-08-30T11:25:09:Creation" ;
```

Please note that for other missions than Sentinel-3 and Sentinel-6 a few global attributes are different:

Institution = CLS
Project = COPERNICUS MARINE ENVIRONMENT MONITORING SERVICE (CMEMS)

VI.2 VI.2 SWH level-4 files (_014_003)

Example of NetCDF metadata from these files is given below.

```
netcdf cmems_obs-wave_glo_phy-swh_nrt_multi_l4_2deg_P1D-m_20220419T120000Z_P20240529T000000Z {
dimensions:
	time = 1 ;
	latitude = 90 ;
	longitude = 180 ;
	nv = 2 ;
variables:
	float time(time) ;
		time:axis = "T" ;
		time:long_name = "time" ;
		time:standard_name = "time" ;
		time:units = "days since 1950-01-01" ;
		time:calendar = "gregorian" ;
	float latitude(latitude) ;
		latitude:axis = "Y" ;
		latitude:bounds = "lat_bnds" ;
		latitude:long_name = "latitude" ;
		latitude:standard_name = "latitude" ;
		latitude:units = "degrees_north" ;
		latitude:valid_max = 89. ;
		latitude:valid_min = -89. ;
	float longitude(longitude) ;
		longitude:axis = "X" ;
		longitude:bounds = "lon_bnds" ;
		longitude:long_name = "longitude" ;
		longitude:standard_name = "longitude" ;
		longitude:units = "degrees_east" ;
		longitude:valid_max = 179. ;
		longitude:valid_min = -179. ;
	int nv(nv) ;
		nv:comment = "Vertex" ;
		nv:units = "1" ;
	float lat_bnds(latitude, nv) ;
		lat_bnds:comment = "latitude values at the north and south bounds of each pixel." ;
	float lon_bnds(longitude, nv) ;
		lon_bnds:comment = "longitude values at the west and east bounds of each pixel." ;
	short VAVH_DAILY_MEAN(time, latitude, longitude) ;
		VAVH_DAILY_MEAN:_FillValue = -32767s ;
		VAVH_DAILY_MEAN:ancillary_variables = "VAVH_DAILY_NBR" ;
		VAVH_DAILY_MEAN:cell_methods = "area: mean" ;
		VAVH_DAILY_MEAN:comment = "Daily statistics are computed from all available altimetry missions Level 3 along-
track measurements from 00 UTC until 23:59 UTC." ;
		VAVH_DAILY_MEAN:coordinates = "longitude latitude" ;
		VAVH_DAILY_MEAN:long_name = "Daily mean significant wave height" ;
		VAVH_DAILY_MEAN:standard_name = "sea_surface_wave_significant_height" ;
		VAVH_DAILY_MEAN:units = "m" ;
		VAVH_DAILY_MEAN:valid_max = 32767 ;
		VAVH_DAILY_MEAN:valid_min = 0 ;
		VAVH_DAILY_MEAN:scale_factor = 0.001 ;
	short VAVH_DAILY_STD(time, latitude, longitude) ;
		VAVH_DAILY_STD:_FillValue = -32767s ;
		VAVH_DAILY_STD:ancillary_variables = "VAVH_DAILY_NBR" ;
		VAVH_DAILY_STD:cell_methods = "area: standard_deviation" ;
		VAVH_DAILY_STD:comment = "Daily statistics are computed from all available altimetry missions Level 3 along-track
measurements from 00 UTC until 23:59 UTC." ;
		VAVH_DAILY_STD:coordinates = "longitude latitude" ;
```

```
        VAVH_DAILY_STD:long_name = "Daily standard deviation of significant wave height" ;
        VAVH_DAILY_STD:standard_name = "sea_surface_wave_significant_height" ;
        VAVH_DAILY_STD:units = "m" ;
        VAVH_DAILY_STD:valid_max = 32767 ;
        VAVH_DAILY_STD:valid_min = 0 ;
        VAVH_DAILY_STD:scale_factor = 0.001 ;
    short VAVH_DAILY_MAX(time, latitude, longitude) ;
        VAVH_DAILY_MAX:_FillValue = -32767s ;
        VAVH_DAILY_MAX:ancillary_variables = "VAVH_DAILY_NBR" ;
        VAVH_DAILY_MAX:cell_methods = "area: maximum" ;
        VAVH_DAILY_MAX:comment = "Daily statistics are computed from all available altimetry missions Level 3 along-
track measurements from 00 UTC until 23:59 UTC." ;
        VAVH_DAILY_MAX:coordinates = "longitude latitude" ;
        VAVH_DAILY_MAX:long_name = "Daily maximum significant wave height" ;
        VAVH_DAILY_MAX:standard_name = "sea_surface_wave_significant_height" ;
        VAVH_DAILY_MAX:units = "m" ;
        VAVH_DAILY_MAX:valid_max = 32767 ;
        VAVH_DAILY_MAX:valid_min = 0 ;
        VAVH_DAILY_MAX:scale_factor = 0.001 ;
    int VAVH_DAILY_NBR(time, latitude, longitude) ;
        VAVH_DAILY_NBR:_FillValue = -999 ;
        VAVH_DAILY_NBR:comment = "Daily statistics are computed from all available altimetry missions Level 3 along-track
measurements from 00 UTC until 23:59 UTC." ;
        VAVH_DAILY_NBR:coordinates = "longitude latitude" ;
        VAVH_DAILY_NBR:long_name = "Daily number of significant wave height measurements" ;
        VAVH_DAILY_NBR:standard_name = "sea_surface_wave_significant_height number_of_observations" ;
        VAVH_DAILY_NBR:units = "1" ;


// global attributes:
        :Conventions = "CF-1.6" ;
        :Metadata_Conventions = "Unidata Dataset Discovery v1.0" ;
        :cdm_data_type = "Grid" ;
        :comment = "Significant Wave Height measured by altimetry" ;
        :contact = "servicedesk.cmems@mercator-ocean.eu" ;
        :creator_email = "servicedesk.cmems@mercator-ocean.eu" ;
        :creator_name = "CMEMS - Wave Thematic Assembly Center" ;
        :creator_url = "http://marine.copernicus.eu" ;
        :date_created = "2024-05-29T00:00:00Z" ;
        :geospatial_lat_max = 89. ;
        :geospatial_lat_min = -89. ;
        :geospatial_lat_resolution = 2 ;
        :geospatial_lat_units = "degrees_north" ;
        :geospatial_lon_max = 179. ;
        :geospatial_lon_min = -179. ;
        :geospatial_lon_resolution = 2 ;
        :geospatial_lon_units = "degrees_east" ;
        :history = "2024-05-29T00:00:00Z: Creation" ;
        :institution = "CLS" ;
        :keywords = "Oceans > Ocean Topography > significant wave height" ;
        :keywords_vocabulary = "NetCDF COARDS Climate and Forecast Standard Names" ;
        :license = "http://marine.copernicus.eu/web/27-service-commitments-and-licence.php" ;
        :platform = "CFOSAT, Cryosat-2, HaiYang-2B, SARAL-Altika, Sentinel-3A, Sentinel-3B, Sentinel-6A" ;
        :processing_level = "L4" ;
        :project = "COPERNICUS MARINE ENVIRONMENT MONITORING SERVICE (CMEMS)" ;
        :references = "http://marine.copernicus.eu" ;
```

```
        :software_version = "L4RT_SWH: 2.0.2" ;
        :source = "Altimetry measurements" ;
        :standard_name_vocabulary = "NetCDF Climate and Forecast (CF) Metadata Convention Standard Name Table v37" ;
        :summary = "Near-Real-Time Level-4 Significant Wave Height and derived variables measured by multi-satellite
altimetry observations over Global Ocean." ;
        :time_coverage_duration = "P1D" ;
        :time_coverage_end = "2022-04-19T12:00:00Z" ;
        :time_coverage_resolution = "P1D" ;
        :time_coverage_start = "2022-04-19T12:00:00Z" ;
        :title = "NRT merged all satellites Global Ocean Gridded Significant Wave Height L4 product and mean variables" ;
        :product_version = "202211" ;
}
```

Note: The instantaneous dataset (cmems_obs-wave_glo_phy-swh_my_multi_l4_2deg_P1D-i) is very similar to the mean dataset (cmems_obs-wave_glo_phy-swh_my_multi_l4_2deg_P1D-m) exposed above. The only difference are: i) variable VAVH_INST instead of VAVH_DAILY_MEAN and ii) variables VAVH_INST_STD, VAVH_INST_MAX and VAVH_INST_NBR does not exist but the flag VAVH_INST_FLAG.

```
}
```

VI.3 SPC OBS CFOSAT level-3 files (_014_009)

Example of instantaneous NetCDF metadata from these files is given below.

```
netcdf global_spc_l3_rep_cfo_20211101T003022Z_ 20211101T020353Z P20220101T020353Z {
dimensions:
    side_looking = 2 ;
    obs = 525 ;
    spectrum_wavenumber = 32 ;
    spectrum_direction = 24 ;
    n_partition = 3 ;
variables:
    double time(side_looking, obs) ;
        time:units = "days since 1950-01-01T00:00:00Z" ;
        time:axis = "T" ;
        time:standard_name = "time" ;
        time:calendar = "gregorian" ;
        time:long_name = "Observation time" ;
    float latitude(side_looking, obs) ;
        latitude:_FillValue = -999.f ;
        latitude:description = "Positive latitude is North latitude, negative value is South latitude." ;
        latitude:units = "degrees_north" ;
        latitude:axis = "Y" ;
        latitude:long_name = "Latitude of observation" ;
        latitude:standard_name = "latitude" ;
        latitude:valid_min = -90.f ;
        latitude:valid_max = 90.f ;
    float longitude(side_looking, obs) ;
```

```
        longitude:_FillValue = -999.f ;
        longitude:description = "East longitude relative to Greenwich meridian" ;
        longitude:units = "degrees_east" ;
        longitude:axis = "X" ;
        longitude:long_name = "Longitude of observation" ;
        longitude:standard_name = "longitude" ;
        longitude:valid_min = -180.f ;
        longitude:valid_max = 180.f ;
    float wavenumber_spec(spectrum_wavenumber) ;
        wavenumber_spec:_FillValue = -999.f ;
        wavenumber_spec:units = "rad/m" ;
        wavenumber_spec:long_name = "Wavenumbers for polar spectra" ;
        wavenumber_spec:standard_name = "wavenumber_spec" ;
        wavenumber_spec:valid_min = 0.f ;
        wavenumber_spec:valid_max = 1.f ;
    float direction_spectrum(spectrum_direction) ;
        direction_spectrum:_FillValue = -999.f ;
        direction_spectrum:units = "degree" ;
        direction_spectrum:long_name = "Array of angular values for polar spectra (degrees from North, clockwise)" ;
        direction_spectrum:standard_name = "direction_spec" ;
        direction_spectrum:valid_min = 0.f ;
        direction_spectrum:valid_max = 360.f ;
    float wave_spec(spectrum_wavenumber, spectrum_direction, side_looking, obs) ;
        wave_spec:_FillValue = -999.f ;
        wave_spec:units = "m^2 / radians" ;
        wave_spec:long_name = "Estimated directional wave slope spectrum  for spectral beam 10 degrees" ;
        wave_spec:standard_name = "wave_spectrum" ;
        wave_spec:valid_min = 0.f ;
        wave_spec:valid_max = 100.f ;
    byte wave_spec_partition_mask(spectrum_wavenumber, spectrum_direction, n_partition, side_looking, obs) ;
        wave_spec_partition_mask:_FillValue = -127b ;
        wave_spec_partition_mask:units = "1" ;
        wave_spec_partition_mask:flag_meanings = "first_part_of_the_partition no_partition
second_part_of_the_partition" ;
        wave_spec_partition_mask:long_name = "Mask of the partitions detected on the wave spectrum (1 and -1 indicate
the partitions and its symmetric counterpart, 0 is outside partition)" ;
        wave_spec_partition_mask:references = "CF-GSFR-SP-804-CNES" ;
        wave_spec_partition_mask:least_significant_digit = 0LL ;
        wave_spec_partition_mask:flag_values = "-1, 0, 1" ;
        wave_spec_partition_mask:valid_min = -1b ;
        wave_spec_partition_mask:valid_max = 1b ;
    float platform_azimuth_angle(obs) ;
        platform_azimuth_angle:_FillValue = -999.f ;
        platform_azimuth_angle:units = "degrees" ;
        platform_azimuth_angle:long_name = "Along track direction" ;
        platform_azimuth_angle:description = "The platform azimuth direction is calculated as the angle between satellite
orbit plane and geographical North" ;
        platform_azimuth_angle:standard_name = "platform_azimuth_angle" ;
        platform_azimuth_angle:valid_min = 0.f ;
        platform_azimuth_angle:valid_max = 360.f ;
    float VAVH(obs, side_looking) ;
        VAVH:_FillValue = -999.f ;
```

```
        VAVH:units = "m" ;
        VAVH:scale_factor = 0.001 ;
        VAVH:long_name = "Significant wave height (SWH)" ;
        VAVH:description = "SWH estimated from whole spectrum." ;
        VAVH:standard_name = "sea_surface_wave_significant_height" ;
        VAVH:valid_min = 0.f ;
        VAVH:valid_max = 2000.f ;
    float VPED(obs, side_looking) ;
        VPED:_FillValue = -999.f ;
        VPED:units = "degree" ;
        VPED:scale_factor = 0.1 ;
        VPED:long_name = "Peak direction (oceano. convention)" ;
        VPED:description = "Peak direction estimated from the whole spectrum. It corresponds to the wave direction (with
180 deg. ambiguity) in degrees from North, clockwise." ;
        VPED:standard_name = "sea_surface_wave_from_direction_at_variance_spectral_density_maximum" ;
    float VTPK(obs, side_looking) ;
        VTPK:_FillValue = -999.f ;
        VTPK:units = "s" ;
        VTPK:scale_factor = 0.01 ;
        VTPK:long_name = "Peak period wavelength" ;
        VTPK:description = "Peak period estimated from the whole spectrum." ;
        VTPK:standard_name = "sea_surface_wave_period_at_variance_spectral_density_maximum" ;
    float VAVH_partition(obs, side_looking, n_partition) ;
        VAVH_partition:_FillValue = -999.f ;
        VAVH_partition:units = "m" ;
        VAVH_partition:scale_factor = 0.001 ;
        VAVH_partition:long_name = "Significant wave height for partition 1, 2, 3" ;
        VAVH_partition:description = "SWH estimated for the three dominant partitions of the spectrum" ;
        VAVH_partition:standard_name = "sea_surface_wave_significant_height_partition_1-2-3" ;
        VAVH_partition:valid_min = 0.f ;
        VAVH_partition:valid_max = 2000.f ;
    float VPED_partition(obs, side_looking, n_partition) ;
        VPED_partition:_FillValue = -999.f ;
        VPED_partition:units = "degree" ;
        VPED_partition:scale_factor = 0.1 ;
        VPED_partition:long_name = "Peak direction for partitions 1, 2, 3" ;
        VPED_partition:description = "Peak direction estimated for the three dominant partitions of the spectrum" ;
        VPED_partition:standard_name = "sea_surface_wave_from_direction_partition_1-2-3" ;
        VPED_partition:valid_min = -180.f ;
        VPED_partition:valid_max = 180.f ;
    float VTPK_partition(obs, side_looking, n_partition) ;
        VTPK_partition:_FillValue = -999.f ;
        VTPK_partition:units = "m" ;
        VTPK_partition:scale_factor = 0.01 ;
        VTPK_partition:long_name = "Peak period for partitions 1,2,3" ;
        VTPK_partition:description = "Peak period estimated for the three dominant partitions of the spectrum." ;
        VTPK_partition:standard_name = "sea_surface_wave_peak_period_partition_1-2-3" ;
        VTPK_partition:valid_min = 0.f ;
        VTPK_partition:valid_max = 1.f ;

// global attributes:
        :L3_Processor_version = "2.6.1" ;
        :L2P_Processor_version = "2.6.1" ;
```

```
        :L2_Processor_version = "OP05" ;
        :source_platform = "Swim measurements" ;
        :license = "http://marine.copernicus.eu/web/27-service-commitments-and-licence.php" ;
        :creator_name = "CMEMS - Wave Thematic Assembly Center" ;
        :title = "NRT Swim Global Ocean Spectral Wave parameters L3 product" ;
        :creator_email = "servicedesk.cmems@mercator-ocean.eu" ;
        :standard_name_vocabulary = "NetCDF Climate and Forecast (CF) Metadata Convention Standard Name Table v37" ;
        :Conventions = "CF-1.6" ;
        :project = "COPERNICUS MARINE ENVIRONMENT MONITORING SERVICE (CMEMS)" ;
        :Keywords_vocabulary = "NetCDF COARDS Climate and Forecast Standard Names" ;
        :contact = "servicedesk.cmems@mercator-ocean.eu" ;
        :references = "http://marine.copernicus.eu" ;
        :based_on = "Swim SW_L2P_OCN" ;
        :creator_url = "http://marine.copernicus.eu" ;
        :institution = "CLS" ;
        :processing_level = "L3" ;
        :creation_date = "2022-10-20T13:53:19" ;
        :comment = "SWIM off-nadir products selected from Beam 10 degrees, and SWIM nadir products averaged by box" ;
        :first_obs_time = "2021-11-01 00:30:26" ;
        :last_obs_time = "2021-11-01 02:03:47" ;
}
```